

EEG acquisition system — data-integrity audit and hardware recommendation

All 84 training recordings, 20 participants, Jun 4 – Aug 12 · OpenBCI Cyton+Daisy rig

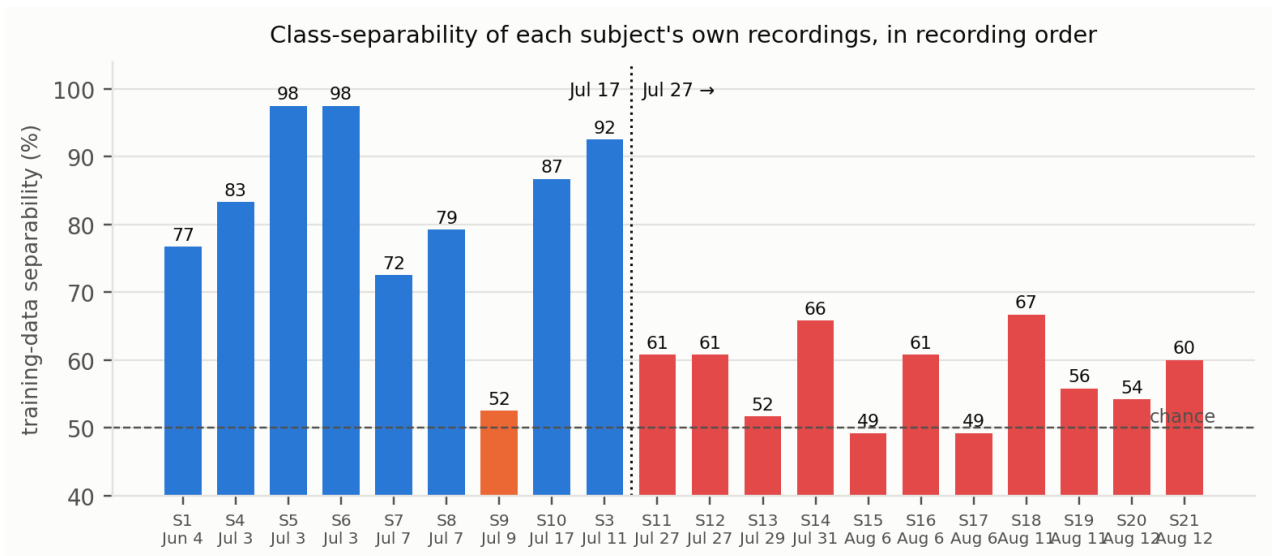
Executive summary

Every training recording in the study was audited for digital integrity (frozen or missing samples, timing), analog quality (noise, amplitudes, referencing), and information content (whether the recording contains a decodable motor-imagery signal at all). The audit finds **two independent failure layers**:

1. The acquisition hardware is failing, progressively and now acutely. Half the montage (the Daisy board) recorded nothing at all for two full sessions in July; on Aug 11-12 the stream froze mid-run and then lost 43-76% of all samples across three consecutive runs. Data recorded during these events is unusable and, worse, looks superficially normal.

2. Since Jul 27, no recording contains a usable motor-imagery signal. Eleven consecutive participants (S11-S21) produced training data at 49-67% class-separability (chance = 50%), against 72-97% for the nine earlier cohort members. The signal is absent at every latency, on every channel — this is why models trained on fresh data fail while models from earlier participants still allow partial control.

Recommendation: **yes, replace the board — and refresh the electrode chain at the same time.** The board's digital failures alone justify replacement, but they do not fully explain the signal loss (the stream was digitally clean for S11-S17 while the signal was already absent), so the reference electrode, gel, and cap placement must be renewed in the same intervention. A 10-minute pre-session check (now scripted) verifies the repair and prevents this class of loss from recurring.

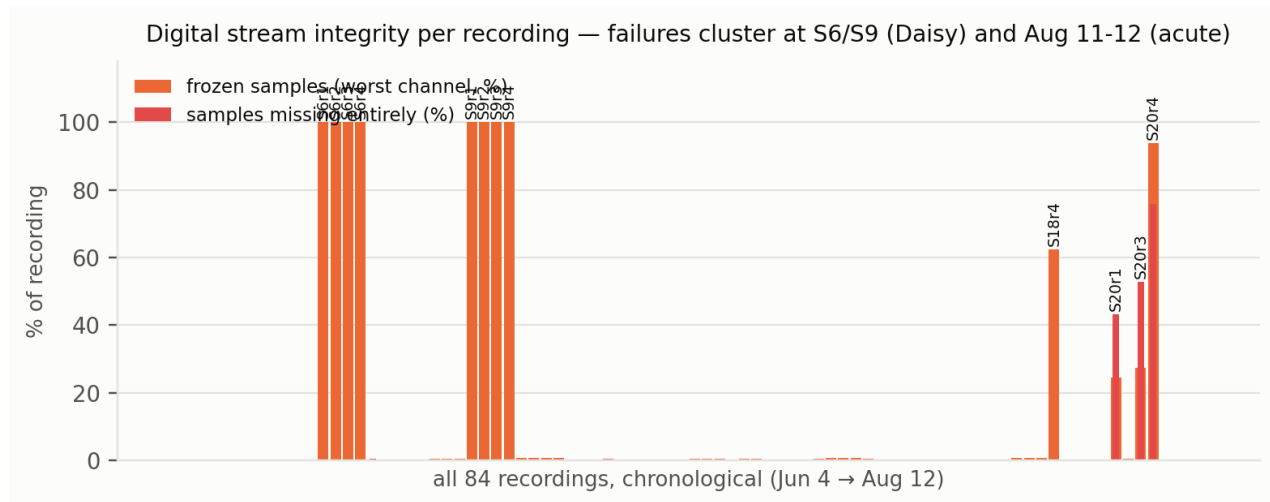


Every participant's own training data, scored by a fixed pipeline (train half / test half). Blue = usable, red = at/near chance, orange = the Daisy-outage session. The break between Jul 17 and Jul 27 is absolute.

1. Hard hardware failures (digital, unambiguous)

Event	When	Evidence	Data impact
Daisy outage — S6	Jul 3	all 8 Daisy channels flat, 100% frozen, whole session	half montage lost ×4 runs
Daisy outage — S9	Jul 9	identical signature	half montage lost ×4 runs; subject unusable
Stream freeze — S18 r4	Aug 11	62% of Cyton samples frozen; amplitudes 5,500–9,300 μV	run unusable
Acute collapse — S20 r1	Aug 12	43% of samples missing; Daisy 24% frozen	run unusable
Acute collapse — S20 r3	Aug 12	53% missing; Daisy 27% frozen	run unusable
Acute collapse — S20 r4	Aug 12	76% missing; Cyton 94% frozen	run unusable

In the S20 runs the saved data timeline is also desynchronized from the cue timeline (cue spacing compressed from 6.33 s to 1.8–3.7 s in the file), so even the surviving samples cannot be correctly labeled. Failure frequency is accelerating: 2 events in Jun–Jul, 4 in the last two days of recording.



Worst-channel frozen-sample fraction and missing-sample fraction for each of the 84 recordings.

2. Signal-content failure (the study-level loss)

For S11–S21 the recordings are digitally clean (0 flat channels, ~0.1% frozen samples) and look like EEG — eye-blinks appear on the correct frontal channels, amplitudes are normal. But the class-discriminative motor signal is absent: no channel exceeds ~0.6 AUC (good era: C3/C4 at 0.6–0.77), and shifting the analysis window anywhere from –4 s to +6 s around the cue finds nothing (good era: sharp peak at the cue, up to 98%). Eleven consecutive chance-level participants after eight consecutive good ones is not participant variability. Software is ruled out: the experiment protocol, cue timing, and analysis code are unchanged across the boundary (verified in version control), and the same pipeline still scores the old recordings at 72–97%.

The remaining causes are physical: reference/ground electrode degradation, electrode-gel contact, cap placement, or analog front-end degradation on the board itself. These cannot be separated from recorded data alone — they are separated by the repair sequence below.

3. Chronic quality issues (both eras, secondary)

Issue	Extent	Reading
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60 Hz line noise escalating within sessions	S8 ×36→83, S17 ×8→78, S16 ×33-42, sporadic ×3-15 elsewhere	ground/bias electrode instability
Very high common-mode correlation (0.7-1.0)	majority of recordings, both eras	weak referencing hygiene
First-run amplitude spikes (600-2,100 μ V), settling by run 2	most participants	gel/contact settling — start recording later
Channel 8 permanently dead (by config)	entire study	montage has run 15/16 channels since June

4. Why old models 'work' but fresh ones do not

Models trained on Jun-Jul data learned genuine motor-imagery patterns; driven by a live signal that is even partially captured, they produce partial control (as observed). Models trained on post-Jul-27 recordings learn noise — there is no signal in their training data — so they cannot work regardless of the online pipeline, thresholds, or calibration. This exactly matches the field observations that prompted this audit.

5. Recommended actions

#	Action	Rationale
1	Stop new participant sessions immediately	12 of 20 participants' data already unusable
2	Replace the Cyton+Daisy board (or RMA); replace dongle batteries as a matter of course	documented progressive digital failure, now acute
3	Replace reference/ground electrode + all gel; verify cap placement (C3/C4 over motor cortex)	board replacement alone does not explain the Jul-27 signal loss
4	Re-validate with scripts/rig_check.py on a known-good pilot (one of S1-S8)	pass = $\geq 75\%$ separability peaked at the cue; 10 minutes
5	Adopt rig_check as a mandatory pre-flight before every session day	would have caught this on Jul 27 instead of 12 participants later
6	Re-record participants S11-S21 once the rig passes	their sessions cannot be salvaged; signal was never captured

Cost framing: the replacement hardware is a small fraction of the value of the twelve participant-sessions already lost (48 recording runs plus staff time). The pre-flight gate makes a recurrence a 10-minute detection instead of a multi-week silent loss.

Prepared from the complete recording archive; every metric in this report is reproducible from the repository (scripts/rig_check.py, scripts/check_board.py, audit scripts in session records).